Demonstration of Fonts and Alignment Filter for Pandoc

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This document demonstrates every feature provided by the `fonts-and-alignment` Lua filter for Pandoc. Each code block shows the exact Markdown syntax used to generate the rendered output that follows, ensuring consistent results across both LaTeX/PDF and HTML formats.[1]

The filter relies on Pandoc's `fenced_divs` and `bracketed_spans` extensions, which are typically enabled by default. In the uncommon event that these extensions are disabled, the Lua filter will be unable to interpret the corresponding divs and spans correctly, causing them to appear as raw text in the rendered output.

Bracketed Spans and Fenced Divs Invocations

The filter may be used with Pandoc's `bracketed_spans` and `fenced_divs` syntax extensions, as demonstrated below. Bracketed Spans are intended for styling smaller inline portions of text, while Fenced Divs are used for styling larger blocks of content. This follows Pandoc's standard syntax conventions. For more information on this syntax, please refer to the official Pandoc documentation: Divs and Spans — Pandoc User's Guide.

```
[This font is extra extra large.]{.pfa-text-2xl}
```

This font is extra extra large.

```
::: {.pfa-text-2xl}
This font is extra extra large.
:::
```

[1] To enable equivalent styling in HTML output, include the `fonts-and-alignment.css` stylesheet distributed with this filter.

This font is extra extra large.

Font Sizing in Bracketed Spans

The filter provides nine predefined sizing hooks that allow text within Bracketed Spans to be scaled relative to the document's base font size.

Size	Syntax	Output
tiny	`[Sample]{.pfa-text-3xs}`	Sample
scriptsize	`[Sample]{.pfa-text-2xs}`	Sample
footnotesize	`[Sample]{.pfa-text-xs}`	Sample
small	`[Sample]{.pfa-text-s}`	Sample
normal	`[Sample]{.pfa-text-normal}`	Sample
large	`[Sample]{.pfa-text-l}`	Sample
Large	`[Sample]{.pfa-text-xl}`	Sample
LARGE	`[Sample]{.pfa-text-2xl}`	Sample
huge	`[Sample]{.pfa-text-3xl}`	Sample

Font Sizing in Fenced Divs

The same sizing classes can also be applied to Fenced Divs, allowing an entire block of text to be rendered at a specific font size.

Extra Small Font Size

```
::: {.pfa-text-xs}
This paragraph renders at the _extra small_ font size.
:::
```

This paragraph renders at the *extra small* font size.

Normal Font Size

```
::: {.pfa-text-normal}
This paragraph renders at the _normal_ font size.
:::
```

This paragraph renders at the *normal* font size.

Large Font Size

```
::: {.pfa-text-l}
This paragraph renders at the _large_ font size.
:::
```

This paragraph renders at the *large* font size.

Font Weights, Shapes, and Families in Bracketed Spans

The following typographic styles can be applied to Bracketed Spans to control weight, style, and typeface.

Style	Syntax	Output
Bold	`[Sample]{.pfa-font-bold}`	**Sample**
Medium	`[Sample]{.pfa-font-medium}`	Sample
Italic	`[Sample]{.pfa-font-italic}`	*Sample*
Slanted	`[Sample]{.pfa-font-slanted}`	*Sample*
Upright	`[Sample]{.pfa-font-upright}`	Sample
Emphasis	`[Sample]{.pfa-font-emphasis}`	*Sample*
Serif	`[Sample]{.pfa-font-serif}`	Sample
Sans-Serif	`[Sample]{.pfa-font-sans}`	Sample
Monospace	`[Sample]{.pfa-font-mono}`	`Sample`
Small Caps	`[Sample]{.pfa-font-smallcaps}`	SAMPLE
Normal	`[Sample]{.pfa-font-normal}`	Sample

Font Weights, Shapes, and Families in Fenced Divs

These same typographic classes can also be applied to Fenced Divs, allowing an entire block of text to adopt a particular visual style.

Bold Weight

```
::: {.pfa-font-bold}
This paragraph renders in bold type.
:::
```

> **This paragraph renders in bold type.**

Sans-Serif Family

```
::: {.pfa-font-sans}
This paragraph uses a sans-serif typeface.
:::
```

> This paragraph uses a sans-serif typeface.

Small Caps

```
::: {.pfa-font-smallcaps}
This paragraph is rendered in small caps.
:::
```

> THIS PARAGRAPH IS RENDERED IN SMALL CAPS.

Text Casing Transformations

Unlike purely visual styling in HTML/CSS—where `text-transform` only changes how text is rendered—these casing transformations operate at the Pandoc Abstract Syntax Tree (AST) level and modify the actual document content. This ensures that uppercase and lowercase conversions are preserved consistently across all output formats and remain intact when copied into other applications.

Casing Transformations in Bracketed Spans

Casing	Syntax	Output
Uppercase	`[Sample]{.pfa-uppercase}`	SAMPLE
Lowercase	`[SAMPLE]{.pfa-lowercase}`	sample

Casing Transformations in Fenced Divs

Lowercase

```
::: {.pfa-lowercase}
THIS PARAGRAPH IS TRANSFORMED TO LOWERCASE.
:::
```

this paragraph is transformed to lowercase.

Uppercase

```
::: {.pfa-uppercase}
this paragraph is transformed to uppercase.
:::
```

THIS PARAGRAPH IS TRANSFORMED TO UPPERCASE.

Text Decorations for Bracketed Spans

Text decoration styles are implemented using the LaTeX `ulem` package, which is automatically included via the `header-includes` field whenever any of the corresponding classes are detected in the document. These styles are also mapped to equivalent CSS properties to ensure consistent rendering across LaTeX/PDF and HTML outputs. These decorations apply only to **Bracketed Spans**.

Decoration	Syntax	Output
Single Underline	`[Sample]{.pfa-text-uline}`	Sample
Double Underline	`[Sample]{.pfa-text-uline-double}`	Sample
Dashed Underline	`[Sample]{.pfa-text-uline-dashed}`	Sample
Dotted Underline	`[Sample]{.pfa-text-uline-dotted}`	Sample
Wavy Underline	`[Sample]{.pfa-text-uline-wave}`	Sample
Strikeout	`[Sample]{.pfa-text-strikeout}`	~~Sample~~

Note: The `\xout{...}` marked-out text style has been removed from this version of the package due to lack of consistent cross-format support in HTML/CSS and limited usage in modern document workflows.

Colors

This section describes how to apply color to text using the `pfa-font-color` attribute in both Bracketed Spans and Fenced Divs.

Solid Colors

The `pfa-font-color` attribute supports CSS3 named colors, full hexadecimal values, and three-digit shorthand hexadecimal values.

Flexible Color Terminology

For standard solid colors, the filter automatically normalizes CSS3 color names. This means solid colors are completely case-insensitive and support various naming conventions. All examples below resolve perfectly to the CSS3 color `mediumvioletred` in both HTML and PDF formats:

Naming Convention	Syntax	Output
Lowercase	`[Color]{pfa-font-color="mediumvioletred"}`	Color
Title Case	`[Color]{pfa-font-color="Medium Violet Red"}`	Color
Kebab Case	`[Color]{pfa-font-color="medium-violet-red"}`	Color
Snake Case	`[Color]{pfa-font-color="medium_violet_red"}`	Color
Camel Case	`[Color]{pfa-font-color="mediumVioletRed"}`	Color
Pascal Case	`[Color]{pfa-font-color="MediumVioletRed"}`	Color
Screaming Snake Case	`[Color]{pfa-font-color="MEDIUM_VIOLET_RED"}`	Color

Applying Solid Colors

Input Type	Syntax	Output
CSS3 Named	`[Sample]{pfa-font-color="crimson"}`	Sample
Hex Full	`[Sample]{pfa-font-color="#2E8B57"}`	Sample
Hex Shorthand[2]	`[Sample]{pfa-font-color="#666"}`	Sample

[2] Three-digit shorthand expands by repeating each hexadecimal digit per RGB channel (e.g., `#666` becomes `#666666`). This only applies when each channel consists of a single repeated hexadecimal digit. Full hexadecimal values without per-channel repetition (e.g., `#2E8B57`) are not eligible for shorthand expansion.

Note: While the default Pandoc LaTeX template loads `x11names` (which includes numbered variants like `LightBlue3`), CSS and web browsers do not recognize these. To ensure your colors render perfectly across both PDF and HTML formats, you must stick strictly to standard CSS3 Named Colors or Hexadecimal codes.

Color Mixing

The filter natively supports LaTeX's `xcolor` percentage mixing syntax, allowing you to tint or shade colors on the fly. This translates perfectly into both PDF outputs and HTML outputs (using the modern CSS `color-mix()` function).

The mixing syntax uses the exclamation mark (`!`) to separate values:

- **Tinting:** `BaseColor!Percentage`. The percentage dictates how much of the base color is kept. (e.g., `Maroon!40` results in 40% Maroon and 60% white).
- **Shading:** `BaseColor!Percentage!black`. By using black as the second color, you darken the base color. (e.g., `MediumVioletRed!80!black` results in 80% MediumVioletRed and 20% black).
- **Mixing Two Colors:** `BaseColor!Percentage!MixColor`. The percentage applies to the first color, and the remaining percentage applies to the second. (e.g., `RoyalBlue!50!ForestGreen` results in 50% RoyalBlue and 50% ForestGreen).

Important Casing Rule: Because mixed colors are passed directly to the LaTeX compiler, the flexible terminology rules do not apply here. You must use the exact casing expected by the LaTeX `xcolor` package, otherwise your PDF generation will fail:

- **Base Colors:** The 19 core LaTeX colors must be strictly lowercase.
- **Extended Web Colors:** The CSS3 named colors must be strictly PascalCase.

Mixing Type	Syntax	Output
Tint (40% Base)	`[Tinted]{pfa-font-color="Maroon!40"}`	Tinted
Shade (80% Base)	`[Shaded]{pfa-font-color="MediumVioletRed!80!black"}`	Shade
Mix (50/50)	`[Mixed]{pfa-font-color="RoyalBlue!50!ForestGreen"}`	Mixed

Inheriting Colors in Fenced Divs

When applied to a Fenced Div, the `pfa-font-color` attribute defines the default text color for the entire block. All enclosed content inherits this color unless explicitly overridden by an inner Bracketed Span.

```
::: {pfa-font-color="DarkSlateGrey"}
The Fenced Div defines _DarkSlateGrey_ as the default text color for this
↪ block.

[This Bracketed Span overrides the inherited color to
↪ _tomato_.]{pfa-font-color="tomato"}

The remaining text continues using _DarkSlateGrey_ for the remainder of
↪ the Div.
:::
```

The Fenced Div defines *DarkSlateGrey* as the default text color for this block.
This Bracketed Span overrides the inherited color to *tomato*.
The remaining text continues using *DarkSlateGrey* for the remainder of the Div.

*__Note:__ The `pfa-font-color` utility is an attribute, not a class, and should not be prefixed with a period (`.`).

Text Alignment within Fenced Divs

The `.pfa-align-*` classes may be used to align text within a Fenced Div. These classes map to LaTeX alignment commands (`\raggedright`, `\centering`, and `\raggedleft`) in PDF output while producing equivalent behavior in HTML.

These classes also preserve explicit line breaks introduced with the backslash (\) character, which is useful for poetry, lyrics, and other text where line structure must be preserved.

Left-aligned Text

```
::: {.pfa-align-left}
This block of text is _left-aligned_.
:::
```

This block of text is *left-aligned*.

Center-aligned Text

```
::: {.pfa-align-center}
This block of text is _center-aligned_.
:::
```

This block of text is *center-aligned.*

Right-aligned Text

```
::: {.pfa-align-right}
This block of text is _right-aligned_.
:::
```

This block of text is *right-aligned.*

Explicit Line Break Preservation

The alignment classes preserve explicit line breaks introduced with the backslash (\) character.

```
::: {.pfa-align-center}
This block of text \
is _center-aligned_ \
while preserving explicit line breaks.
:::
```

This block of text
is *center-aligned*
while preserving explicit line breaks.

Block-Level Alignment

The `.pfa-block-*` utilities control the horizontal positioning of a Fenced Div as a whole, without affecting the internal text alignment of its contents. This allows the block itself to be positioned independently of how text is arranged within it.

Left-Aligned Block

```
::: {.pfa-block-left}
The entire block is _left-aligned_.
:::
```

The entire block is *left-aligned*.

Center-Aligned Block

```
::: {.pfa-block-center}
The entire block is _center-aligned_.
:::
```

The entire block is *center-aligned*.

Right-Aligned Block

```
::: {.pfa-block-right}
The entire block is _right-aligned_.
:::
```

The entire block is *right-aligned*.

Combining Multiple Classes

Multiple typographic utilities can be combined within the same element. Font, size, color, alignment, and decoration classes are designed to compose independently and can be applied together to both Bracketed Spans and Fenced Divs.

Class names are space-separated within `{}` following Pandoc attribute syntax. In addition to classes, key-value attributes such as `pfa-font-color` may be included alongside class definitions.

Bracketed Span Composition

Style	Syntax	Output
Bold Sans-Serif	`[Sample]{.pfa-font-bold .pfa-font-sans .pfa-text-l pfa-font-color="red"}`	**Sample**
Italic Monospace	`[Sample]{.pfa-font-italic .pfa-font-mono .pfa-text-s}`	*Sample*
Small Caps, Underlined, Colored	`[Sample]{.pfa-font-smallcaps .pfa-text-uline pfa-font-color="forestgreen"}`	SAMPLE

Fenced Div Composition

Multiple utilities can also be applied to Fenced Divs to control alignment, typography, and color simultaneously. The resulting block inherits all specified styles while preserving Pandoc's standard attribute behavior.

```
::: {.pfa-align-center .pfa-font-sans .pfa-font-bold .pfa-text-l
 ↪  pfa-font-color="midnightblue"}
A centered, bold, sans-serif, large, midnight-blue Fenced Div
 ↪  demonstrating multiple combined utilities from the filter.
:::
```

A centered, bold, sans-serif, large, midnight-blue Fenced Div demonstrating multiple combined utilities from the filter.

Legacy Aliases (Deprecated)

Warning: Legacy aliases are deprecated. They are retained strictly for backward compatibility and will be entirely removed in the next major release. New documents should use the `.pfa-*` namespaces going forward.

Font Sizing Aliases

Legacy Classes	Modern Classes
`.xsmall`	`.pfa-text-xs`
`.small`	`.pfa-text-s`
`.normal`	`.pfa-text-normal`
`.large`	`.pfa-text-l`

Legacy Classes	Modern Classes
.xlarge	.pfa-text-xl
.xxlarge	.pfa-text-2xl
.huge	.pfa-text-3xl

Font Weight, Shape, and Family Aliases

Legacy Classes	Shorthand	Modern Classes
.bold	.bf	.pfa-font-bold
.emphasis	.em	.pfa-font-emphasis
.italic	.it	.pfa-font-italic
.medium	.md	.pfa-font-medium
.monospace	.tt	.pfa-font-mono
.normalfont	.nf	.pfa-font-normal
.sans	.sf	.pfa-font-sans
.serif	.rm	.pfa-font-serif
.slanted	.sl	.pfa-font-slanted
.smallcaps	.sc	.pfa-font-smallcaps
.upright	.up	.pfa-font-upright

Alignment Aliases

Legacy Classes	Modern Classes
.centering	.pfa-align-center
.raggedleft	.pfa-align-right
.raggedright	.pfa-align-left

Text Decoration Aliases

Legacy Classes	Shorthand	Modern Classes
.uline	.u	.pfa-text-uline
.uuline	.uu	.pfa-text-uline-double
.dashuline	.dau	.pfa-text-uline-dashed
.dotuline	.dou	.pfa-text-uline-dotted
.uwave	.uw	.pfa-text-uline-wave
.sout	.so	.pfa-text-strikeout

Removed Classes

The following legacy classes were removed in version 2.0.0 and are no longer recognized by the filter. Existing documents should be updated to use the replacement classes shown below.

Removed Class	Shorthand	Removed In	Replacement
`.center`	—	2.0.0	`.pfa-align-center`
`.flushleft`	—	2.0.0	`.pfa-align-left`
`.flushright`	—	2.0.0	`.pfa-align-right`
`.xout`	`.xo`	2.0.0	`.pfa-text-strikeout`